

Pla-tsugi: Terrain as Fracture

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Abstract

Pla-tsugi: Terrain as Fracture reimagines fracture not as loss but as a generative condition. A stone collected from the Rokko Mountain Range (Kobe, Japan) is regarded not as an isolated object but as a fragment once belonging to a continuous terrain. Through 3D scanning, the stone's geometry is digitally aligned with topographic data from Japan's Geospatial Information Authority [Geospatial Information Authority of Japan (GSI), 2024], repositioning the fragment within a larger geological context. Mixed Reality (MR) tools—Meta Quest 3 and Gravity Sketch—were employed to design connective forms conceived as poetic sutures rather than structural joints, emphasizing gestures of care over closure. Materialized through PLA-based 3D printing, these sutures merge with the stone, producing hybrid works where natural matter and digital reconstruction intersect. In exhibitions, viewers are invited to cradle palm-sized fragments, transforming spectatorship into intimate participation. For international presentations, where transporting the original stone is not possible, high-resolution replicas are used to preserve tactile immediacy. This work belongs to the “Pla-tsugi” series, a neologism combining “plastic” and kintsugi (the Japanese art of ceramic repair with gold) [Harada, 2024]. It extends the Japanese repair philosophy from vessels to environments, proposing repair as an act of reconnection between humans, materials, and landscapes.

CCS Concepts

• Human-centered computing; • Human computer interaction (HCI) Applied computing; • Arts and humanities;

Keywords

Pla-tsugi, Mixed Reality, Digital Fabrication, Japanese Repair Philosophy, Terrain, Generative Futures

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1 Introduction

Traditional kintsugi highlights cracks in ceramics with gold, treating rupture as an aesthetic and philosophical opportunity. This project extends that concept to landscapes, reframing fracture as

an opening for renewal rather than an end. By merging fragments, environmental data, and digital tools, Pla-tsugi: Terrain as Fracture builds upon previous research on digital fabrication and contemporary kintsugi practices (Geospatial Information Authority of Japan [Geospatial Information Authority of Japan (GSI), 2024] [Harada, 2024]. It explores how cultural philosophies of repair can resonate with emerging technologies and reconfigure the relationship between humans, materials, and terrain.

2 METHODS AND PROCESS

The workflow begins with 3D scanning a stone and aligning its geometry with GSI terrain data, situating the fragment within a temporal and spatial continuum. MR tools are then used to design sutures that metaphorically “heal” the fracture, emphasizing poetic reconnection. These sutures are materialized through PLA-based 3D printing, producing hybrid forms that merge natural fragments with digital reconstructions. Figure 1 shows the overall workflow of this process.

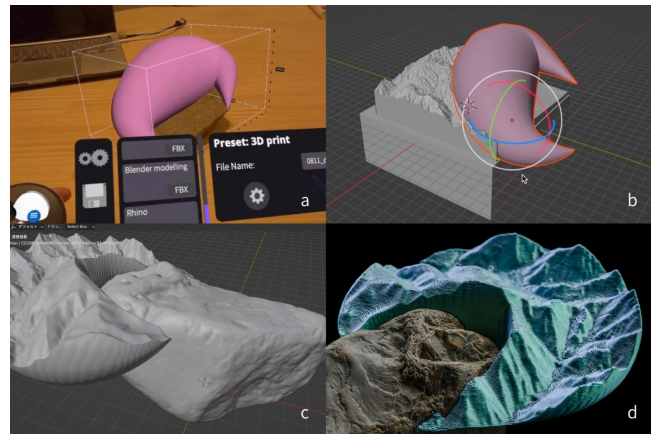


Figure 1: Workflow — (a) MR capture, (b) terrain alignment, (c) merged reconstruction, (d) PLA suture integrated with the original stone. The final artifact is a PLA-based hybrid form that fuses geological history with computational reconstruction.

3 DISCUSSION

Exhibition design emphasizes tactile and participatory engagement. Viewers cradle the fragment in their hands, transforming repair into an embodied act of cherishing (see Figure 2). For international exhibitions where the stone cannot be transported, high-resolution replicas ensure immediacy is preserved.

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Figure 2: Viewing the work in the hand. Exhibited at Rokko Meets Art 2024 beyond (Kobe, Japan).

A supplementary video will accompany this paper at the SIGGRAPH Asia 2025 Art Gallery presentation.

The video documents the conceptual and technical process of creating *Pla-tsugi: Terrain as Fracture*, including 3D scanning, MR-based modeling, and PLA-based fabrication combined with natural stones.

It serves as a visual explanation that complements the written description of the artwork and provides viewers with a clearer understanding of the creative methodology.

4 CONCLUSION

By linking Japanese repair philosophy with XR and 3D printing, Pla-tsugi: Terrain as Fracture reframes rupture as a generative condition. The work resonates with the SIGGRAPH Asia 2025 Art Gallery theme, Generative Futures, positioning repair as both a cultural practice and a sustainable artistic methodology.

References

- Geospatial Information Authority of Japan (GSI). 2024. Digital elevation map data (STL format) of the Rokko Mountain Range, Kobe, Japan, slightly vertically exaggerated to enhance the topographical representation. Available at: <https://maps.gsi.go.jp>
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